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CONFIGURABLE DISPLAY HOLDING APPARATUS AND SYSTEM

Abstract

A display holding apparatus includes a base, a vertical member, and a bracket. The vertical member is attachable to the base in one of two different orientations to form a stand. The bracket is attachable to the vertical member of the stand in one of four different orientations. The four different bracket orientations includes a low landscape orientation, a high landscape orientation, a low portrait orientation, and a high portrait orientation. A display is attachable to the bracket in one of the four bracket orientations. In an embodiment, the base is mounted on a top surface of a transaction terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/103,650, filed Jan. 31, 2023, which application and publication is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Displays and/or illuminated signs come in all a variety of different shapes, sizes, and capabilities. Displays/signs are also popular for advertisements or for drawing a customer's attention to another device, such as an automated teller machine (ATM). Any a sign/display affixed to a top surface of an ATM is often referred to as a “topper.”

[0003] Current designs of ATM toppers are bulky, heavy, and inflexible. There exists just a few design options and each option is restricted to a designated position and orientation on the top surface of the ATM. Existing toppers are also time consuming to assemble and are usually shipped to a retail site separately from the ATM.

SUMMARY

[0004] In various embodiments, a configurable display holding apparatus and system are presented. The apparatus includes a base, a vertical member, and a bracket. The vertical member is attached to the base to form a stand. The bracket attaches to the vertical member of the stand in a plurality of orientations. A display attaches to the bracket in the bracket's configured orientation with the vertical member of the stand.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram of a stand for a display holding apparatus configured in a back orientation, according to an example embodiment.

[0006] FIG. 2 is a diagram of the stand for a display holding apparatus configured in a front orientation, according to an example embodiment.

[0007] FIG. 3 is a diagram of a base for the stand of the display holding apparatus, according to an example embodiment.

[0008] FIG. 4 is a diagram of a configurable bracket for attaching to the stand of the display holding apparatus, according to an example embodiment.

[0009] FIG. 5 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a high portrait orientation, according to an example embodiment.

[0010] FIG. 6 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a low portrait orientation, according to an example embodiment.

[0011] FIG. 7 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a low landscape orientation, according to an example embodiment.

[0012] FIG. 8 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a high landscape orientation, according to an example embodiment.

[0013] FIG. 9 are diagrams illustrating the bracket attached to the stand for four different display orientations, according to an example embodiment.

[0014] FIG. 10 are diagrams illustrating different sized displays held by the display holding apparatus, according to an example embodiment.

[0015] FIG. **11** are diagrams of transaction terminals with displays held by the display holding apparatus, according to an example embodiment.

[0016] FIG. **12** is another diagram of a transaction terminal with a display held by the display holding apparatus, according to an example embodiment.

[0017] FIG. **13** is a block diagram of a system that includes the display holding apparatus, according to an example embodiment.

DETAILED DESCRIPTION

[0018] As stated above, most transaction terminals, such as ATMs, include a topper affixed to a top surface of the ATMs for purposes of providing customer advertisements to customers passing by and/or for purposes of enticing customers to perform transactions on the ATMs. In most cases, these toppers are just lighted signs with fixed and rigid designs attached to the top surface of the ATMs. The dimensions, orientations, and position of the toppers on the ATM are not configurable. More often than not, the retailer selects a topper design and then has a sign panel custom made in the size and dimensions supported by the topper design; a light behind or in the panel provides the retailer's advertisement. Consequently, many toppers appear out of place with the overall aesthetic design of the ATM or the surroundings of where the ATM is located.

[0019] Customers frequent transaction terminals to perform transactions, such as to purchase an item, withdraw currency, deposit currency, place an order with a retailer, review a status of an order, make a reservation, check-in to an establishment, etc. Accordingly, the attention of a customer is captured during a transaction at a terminal. This makes for prime real estate for retailers to pitch their products and services or to establish a retailer's brand with the customer. Additionally, displays of the terminals are often intentionally obfuscated to passers-by so that a customer performing a transaction has privacy; thus, the space above the terminal display can also be used to attract a customer to the terminal for a transaction.

[0020] The teachings provided address these issues by providing a configurable display holder apparatus and system. The apparatus can be configured to hold a lighted panel display or an intelligent display in eight different configurations, four for portrait display orientation and four for landscape display orientation. The apparatus includes a minimal amount of components and can be assembled in minutes. Moreover, the apparatus can be reconfigured for a different display orientation from an existing display orientation in minutes. The components of the apparatus are less than existing components of existing industry toppers; weigh less resulting in reduced shipping expenses; and can be setup and/or reconfigured without any skilled technician in minutes.

[0021] As used herein a “topper” is a device that includes panels which are lighted to produce a visual branding or advertisement on top of a transaction terminal. A “transaction terminal” is a self-service terminal (SST), an automated teller machine (ATM), or a kiosk. A “display” can include a topper with lighted panels to visually convey a message or a branding, a monitor, a digital sign, or an independent processing device integrated into the monitor (e.g., a tablet, touch-based monitor, smart/intelligent monitor/display, etc.).

[0022] The display apparatus **160** (shown in FIG. **10** below) includes a stand **125** (shown in FIG. **1** below), a bracket **130**, a bracket and display cover **140** (shown in FIG. **10** below), and optionally a display **200** (shown in FIG. **10** below). FIG. **1** is a diagram **100** of a stand **125** of a display holding apparatus **160** configured in a back orientation, according to an example embodiment. FIG. **2** is a diagram **200** of the stand **125** of a display holding apparatus **160** configured in a front orientation, according to an example embodiment. FIG. **3** is a diagram **300** of a base **110** for the stand **125** of the display holding apparatus **160**, according to an example embodiment.

[0023] The base **110** includes six premanufactured screw holes or apertures **111**, **112**, and **113**. The back orientation shown in FIG. **1** utilizes screw holes **111** and **112** for purposes of securing a vertical member **120** to the base **110** to form the stand **125**. The front orientation shown in FIG. **2** utilizes screw holes **112** and **113** for purposes of securing vertical member **120** to base **110** to form the stand **135**. The vertical member **120** attaches to the base **110** at a right angle to the base **110**.

Screw holes **112** are common to both the back and front orientations of the stand **125**. Corresponding screw holes **121** are shown at the bottom of vertical member **120** in FIG. 5 below. [0024] The four screw holes **121** of vertical member **120** only align with four of the six screw holes **111**, **112**, and **113**. This ensures that during setup the vertical member **120** can be easily aligned with base **110** and mounted via 4 screws as stand **125** to a top surface of a terminal. The four aligned screw holes **121** form a back orientation when aligned to **111** and **112** of base **110**. A back orientation is one in which the vertical member **120** sits farther away from the front-edge of the terminal. The four aligned screw holes **121** form a front orientation when aligned to **112** and **113** of base **110**. The front orientation is one in which the vertical member **120** sits closer to the front-edge of the terminal.

[0025] FIG. 4 is a diagram **400** of a configurable bracket **130** for attaching to the stand of **125** the display holding apparatus **160**, according to an example embodiment. Bracket **130** includes a plurality of manufactured tabs and screw holes for affixing the bracket **130** to stand **120** and movable plugs for affixing a display **200** (shown in FIG. 10 below). Bracket **130** can be configured to orient an attached display in four different orientations: a low portrait orientation (shown in FIGS. 6 and 9 below), a high portrait orientation (shown in FIGS. 5 and 9 below), a low landscape orientation (shown in FIGS. 7 and 9 below), and a high landscape orientation (shown in FIGS. 8 and 9 below). The bracket **130** adapted to attach to the vertical member **120** in parallel with the orientation of the vertical member **120** or at a right angle to the vertical member **120**. The in parallel attachment of the bracket **130** relative to the vertical member **120** is shown in FIGS. 5 and 6 and the bottom two diagrams in FIG. 9. The right-angle attachment of the bracket **130** relative to the vertical member **120** is shown in FIGS. 7 and 8 and the top two diagrams in FIG. 9.

[0026] FIG. 5 is a diagram **500** illustrating the tab members **135** of the configurable bracket **130** to attach the bracket to the vertical member **120** of the stand **125** for a display **200** in a high portrait orientation, according to an example embodiment. Vertical member **120** includes manufactured slots **122**, which tabs **135** slide of bracket **130** slide into to configure bracket **130** for holding a display **200** in a high portrait orientation.

[0027] Tabs **134** are inserted into slots **122** to configure bracket **130** for holding a display **100** in a low portrait orientation. Bracket **130** can also be rotated 90 degrees relative to vertical member **120** and tabs **132** and **133** inserted into slots **122** for holding a display in high or low landscape orientation. Plungers **131** pull out and include a protective rubber tip for inserting edges a display **200** and/or for inserting a back surface of display **200** against bracket **130**. When the plungers **131** are released, they push into the sides/back surface of display **200** and hold display **200** firmly against bracket **130**.

[0028] FIG. 6 is a diagram **600** illustrating the tab members of the configurable bracket **130** to attach the bracket **130** to the stand **125** for a display **200** in a low portrait orientation, according to an example embodiment. Tabs **134** are inserted into slots **122** with tabs **135** beneath tabs **134**. Tabs **135** fit through other slots **122** such that the bottom of tabs **134** rest on top of vertical member **120** in top slots **122**. This lowers bracket **130** below a vertical height of bracket **130** from that which was shown in FIG. 5 for the high portrait orientation. In the high portrait orientation as shown in FIG. 5 tabs **134** are above tabs **135** which rest on a top of vertical member **120** elevating bracket **130** to a higher vertical position than what is shown in FIG. 6.

[0029] FIG. 7 is a diagram **700** illustrating the attachment tab members of the configurable bracket **130** to attach the bracket **130** to the stand **125** for a display **200** in a low landscape orientation, according to an example embodiment. Bracket **130** is rotated 90 degrees counter clockwise relative to vertical member **120**. Tabs **133** are inserted into top slots **122** of vertical member **120** to configure a low landscape orientation for any subsequently attached display **200**.

[0030] FIG. 8 is a diagram **800** illustrating the tab members of the configurable bracket **130** to attach the bracket **130** to the stand **125** for a display **200** in a high portrait orientation, according to an example embodiment. Once again bracket **130** is rotated 90 degrees counter clockwise relative

to vertical member **120**. Tabs **132** are inserted into top slots **122** of vertical member **120** and rests on top of vertical member **120**. Notice tabs **133** sit vertically above tabs **132** and are not inserted into any tabs in vertical member **120** this provides a higher vertical position for bracket **130** relative to the vertical member **120** that what was shown in FIG. 7 for the low landscape orientation. When tabs **132** are inserted into top slots **122** of vertical member **120** any subsequently attached display **200** is in a high landscape orientation.

[0031] FIG. 9 are diagrams **800** illustrating the bracket **130** attached to the stand **125** on vertical member **120** for four different display orientations, according to an example embodiment. The top rightmost figure shows a low landscape orientation of display apparatus **160** and is configured in the manner discussed above with FIG. 7. The top leftmost figure shows a high landscaper orientation of display apparatus **160** and is configured in the manner discussed above with FIG. 8. The bottom rightmost figure shows a low portrait orientation of display apparatus **160** and is configured in the manner discussed above with FIG. 6. The bottom leftmost figure shows a high portrait orientation of display apparatus **160** and is configured in the manner discussed above with FIG. 5.

[0032] FIG. 10 are diagrams **1000** illustrating different sized displays **200** held by the display holding apparatus **160**, according to an example embodiment. An optional display bezel or cover **140** is shown that fits around a periphery of the display **200** attached to bracket **130** of display apparatus **160**. Diagrams **1000** also illustrate a top tab **137** of bracket **130** under which the display **200** is tucked under. Plungers **131** are pulled out and released causing plungers **131** to push against and hold the sides of display **200** against bracket **130**. Cover **140** can optionally be placed around the periphery of the display **200**. The diagrams **1000** of FIG. 10 illustrate the display apparatus **160** configured in a back and low portrait orientation.

[0033] A desired display **200** is attached to display holding apparatus **160** with a power and/or power and data cord **150** from lights, light emitting diodes (LEDs) board, and/or data port of display **200** running underneath and through base **110** of display holding apparatus **160**. This allows the power or combined power and data wire **150** to be connected accordingly to the terminal without any visible cord **150** when a bottom side of base **110** is mounted to a top side surface of the terminal.

[0034] It is noted that for each of the four discussed display apparatus orientations. The display apparatus **160** can also be in a front orientation (i.e. front edge of stand **120** is closer to the front edge of the terminal) as shown in FIG. 2 or display apparatus **160** can be back orientation (e.g., front edge of stand **120** is farther from the front edge of the terminal) as shown in FIG. 1. Thus, there are 8 possible configurations achievable with display apparatus **160** via just three components for base **110**, stand **120**, and bracket **130** or via four components for optional display cover **140**.

[0035] In an embodiment, four of the six screw holes **111**, **112**, and **113** in base **110** are manufactured to align with existing screw receptacles (**4**) in a top surface of a transaction terminal. Screw holes **112** align with at least two of the four existing screw receptacles and whatever orientation front or back selected by a customer for the display holding apparatus **160** ensures that at least two of screw holes **111** or two of screw holes **113** align with the remaining two existing screw receptacles. Thus, apparatus **160** can be used to replace an existing topper of a terminal with just 3 screws and a desired configuration of eight potential configurations setup without any screws or any tools being required.

[0036] In an embodiment, the display holding apparatus **120** includes an LED board and a panel that sits in front of the LED board to form a display **200** through control of the LEDs on the board illuminated through the panel. In an embodiment, display holding apparatus is sized to hold a tablet touch computer display **200**.

[0037] In an embodiment, bracket **130** is resizable to hold different sized displays **200**. For example a central rail running horizontally across the bracket **130** and a central rail or pair of side rails running vertically across the bracket can be expand out to increase the size of the bracket

horizontally and vertically or fold/tuck in to decrease the size of the bracket horizontally and vertically. In this way, display holding apparatus **160** can be custom sized to accommodate different sized displays **200**.

[0038] FIG. **11** are diagrams **1100** of transaction terminals **300** with displays held by the display holding apparatus **160**, according to an example embodiment. The leftmost figure shows the display apparatus **160-1** configured in a low portrait orientation for the display **200** and a back orientation for stand **120**. The rightmost figure shows the display apparatus **160-2** configured in a high portrait orientation for the display **200** and a front orientation for stand **120**. Although not shown, there can also be a low portrait orientation in a front orientation for the stand and a high portrait orientation in a back orientation from stand **120**.

[0039] FIG. **12** is another diagram **1200** of a transaction terminal **300** with a display **200** held by the display holding apparatus **160**, according to an example embodiment. The display **200** is in a high landscape orientation and a back orientation for the stand **120** on top of terminal **300**.

Although not shown, there can also be a high landscape orientation for display **200** in a front orientation for the stand **120**, a low landscape orientation for display **200** in a front orientation for the stand **120**, and a low landscape orientation for display **200** in a back orientation for the stand **120**.

[0040] Again, there can be any of four different display orientations and each of the four can also be in either a front orientation for the stand **120** or a back orientation for the stand **120**. Thus, there are eight different configurations achievable with display apparatus **160**.

[0041] FIG. **13** is a block diagram of a system **1300** that includes the display holding apparatus **160**, according to an example embodiment. It is noted that system **1300** is shown in greatly simplified form with only those components necessary for understanding the embodiments provided. Thus, there can be more or less components without departing from the teachings provided herein.

[0042] System **1300** includes a transaction terminal **300**, display holding apparatus **160**, and display **200**. The display holding apparatus **160** includes a base **110**, a vertical member **120**, and a bracket **130**. In an embodiment, display holding apparatus **160** includes a display cover **140**. In an embodiment, display holding apparatus **160** includes an LED light board and a light panel. In an embodiment, the display **200** is a lighted sign with lights and one or more covers/panels, a monitor, a touch monitor, or a tablet.

[0043] The vertical member **120** is attachable to base **110** in a front orientation or a back orientation. The bracket **130** is attachable to the vertical member **120** in a high landscape orientation, a low landscape orientation, a high portrait orientation, or a low portrait orientation. Thus, display holding apparatus **160** can be configured in 8 different configurations.

[0044] In an embodiment, the apparatus **160** when holding a specific display **200** forms a display system.

[0045] The above description is illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of embodiments should therefore be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

[0046] In the foregoing description of the embodiments, various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting that the claimed embodiments have more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Description of the Embodiments, with each claim standing on its own as a separate exemplary embodiment.

Claims

1. (canceled)
2. A method, comprising: receiving an electromagnetic noise signal on a first read head of a card reader; receiving the electromagnetic noise signal on a second read head of the card reader; transforming electromagnetic waves associated with the electromagnetic noise signal on the second read head and applying to the first read head to cancel the electromagnetic noise signal on the first read head; and reading magnetic encoded information on a magnetic strip of a card inserted into a card slot of the card reader while preventing any skimmer from reading the magnetic encoded information on the magnetic strip because of a presence of the electromagnetic noise signal.
3. The method of claim 2, wherein transforming further includes connecting the second read head to the first read head in antiphase to achieve the transforming.
4. The method of claim 2, wherein receiving the electromagnetic noise signal on the first read head further includes receiving the electromagnetic noise signal from a disruptive electromagnetic circuit of the card reader.
5. The method of claim 2, wherein receiving the electromagnetic noise signal on the first read head further includes receiving the electromagnetic noise signal from a disruptive electromagnetic source generator that is external to the card reader.
6. The method of claim 2, wherein reading further includes aligning the first read head within the card slot to be situated under track 2 of the magnetic strip of the card.
7. The method of claim 2, wherein reading further includes elevating the first read head at a height above a top of the second read head.
8. The method of claim 7, wherein elevating further includes setting the height to approximately 1 to 2 millimeters.
9. The method of claim 2, wherein reading further includes connecting a first terminal and a second terminal together through the first read head.
10. The method of claim 9, further comprising connecting a third terminal and a fourth terminal together through the second read head.
11. The method of claim 10, further comprising connecting the third terminal to the first terminal and connecting the fourth terminal to the second terminal.
12. The method of claim 2, wherein reading further includes reading the magnetic encoded information through a motorized card reader as the card reader.
13. A method, comprising: receiving a card within a card slot of a terminal; activating an electromagnetic noise signal within the terminal; providing the electromagnetic noise signal to a card reader integrated into the terminal; receiving magnetic encoded information from the card reader after the card reader processes the electromagnetic noise signal to cancel effects of the electromagnetic noise signal on a first read head while maintaining the electromagnetic noise signal on a second read head; and processing a transaction using the magnetic encoded information.
14. The method of claim 13, wherein activating further includes generating the electromagnetic noise signal from a disruptive electromagnetic source generator of the terminal.
15. The method of claim 13, wherein providing further includes sending the electromagnetic noise signal to both the first read head and the second read head of the card reader.
16. The method of claim 13, wherein receiving further includes obtaining the magnetic encoded information from track 2 of a magnetic strip of the card.
17. The method of claim 13, wherein processing further includes processing the transaction as an automated teller machine (ATM) transaction.
18. The method of claim 13, wherein processing further includes processing the transaction as a point-of-sale (POS) transaction.
19. The method of claim 13, wherein processing includes processing the transaction as a self-

service terminal (SST) transaction.

20. A system, comprising: a card reader comprising: a first read head; a second read head connected to the first read head; wherein the first read head and second read head are configured to: receive an electromagnetic noise signal, transform electromagnetic waves associated with the electromagnetic noise signal received on the second read head, apply the transformed electromagnetic waves to the first read head to cancel the electromagnetic noise signal on the first read head, and permit the first read head to read a magnetic strip of a card while the electromagnetic noise signal remains active on any skimmer present within the card reader.

21. The system of claim 20, wherein the first read head is elevated at a height above the second read head and aligned under track 2 of the magnetic strip when the card is inserted into a card slot of the card reader.
